

Multi-Tier Computing -19Z011

Enterprise Information Sharing System (EISS) -
Presentation Tier, Application Tier and Data Tier

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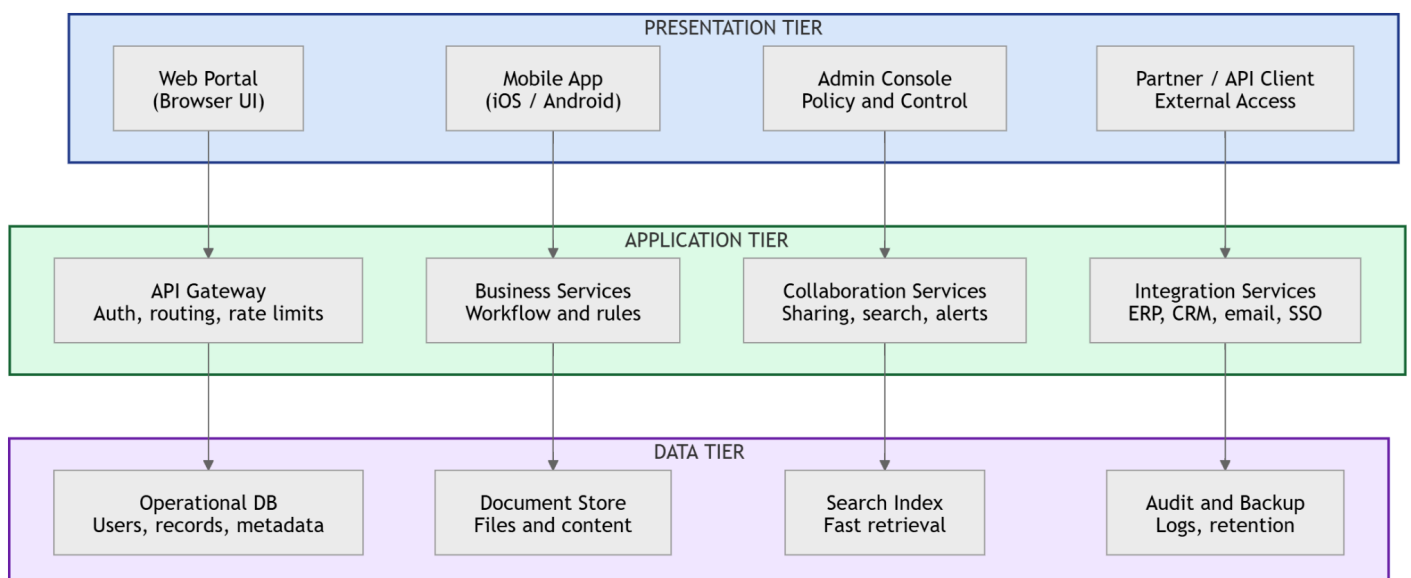
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Introduction:

An enterprise information sharing system is a platform that enables employees, managers, departments, and external stakeholders to create, exchange, search, update, and govern information across the organization. In practice, such a system can support document management, knowledge sharing, workflow approvals, reporting, partner collaboration, case tracking, and operational decision-making.

A robust enterprise solution usually adopts a three-tier architecture. The presentation tier manages how users interact with the system. The application tier implements business logic, security rules, workflow control, and integration. The data tier stores enterprise information, preserves integrity, supports retrieval, and maintains history. Separating these responsibilities reduces complexity, improves security, and allows each tier to scale independently.

This report explains the functions of the presentation tier, application tier, and data tier in detail; shows how requests move through the system; discusses security, performance, and governance considerations; and provides practical guidance for designing and operating an enterprise information sharing system.



Three Tier architecture of a sample enterprise system

Advantages of Three Tier Architecture:

Reason	Explanation
Separation of concerns	Each tier focuses on a clear responsibility, which makes the system easier to design, test, improve, and troubleshoot.
Scalability	Web servers, application servers, and databases can be scaled independently depending on the demand profile.
Security	Sensitive data is not exposed directly to end users; requests must pass through controlled business and access layers.
Maintainability	User interface changes can often be made without redesigning the database, and data structures can evolve without rebuilding the entire front end.
Reusability and integration	Business services can be reused by multiple channels such as web portals, mobile apps, APIs, and partner systems.

A multi-tier system improves organization and flexibility by separating responsibilities across layers. This makes it easier to scale different parts independently, enhance security by controlling access to data, and maintain or update components without affecting the whole system. Additionally, it supports reusability, allowing the same business logic to serve multiple platforms like web, mobile, and APIs.

Presentation Tier:

The **Presentation Tier** is the topmost layer of an Enterprise Information Sharing System. It is responsible for **user interaction and communication with the system**. This layer provides the interface through which users access, input, and visualize shared enterprise data.

In an EISS, the presentation tier ensures that **information is delivered in a clear, accessible, and role-specific manner** to different stakeholders such as employees, administrators, and external partners.

Key Functions

- **User Interface Rendering**

- The presentation tier displays enterprise data using graphical interfaces such as dashboards, forms, and reports. It ensures that complex organizational data is presented in a **user-friendly and understandable format**.

- **User Input Handling**

- It captures user inputs such as queries, file uploads, and form submissions. These inputs are then forwarded to the application tier for processing.

- **Client-Side Validation**

- Basic validation checks (e.g., required fields, format validation) are performed before sending data to reduce unnecessary processing at the backend.

- **Communication with Application Tier**

- The presentation layer communicates with backend services via APIs (REST/GraphQL). It sends user requests and receives processed responses.

- **Personalization and Role-Based Views**

- In an enterprise environment, different users require different views of data. The presentation tier supports **role-based access**, ensuring users see only relevant information.

Application Tier:

The **Application Tier** is the core layer of the Enterprise Information Sharing System. It is responsible for **processing business logic, managing workflows, and coordinating communication between the presentation and data tiers.**

This layer ensures that enterprise data is **processed, validated, and shared according to organizational rules and policies.**

Key Functions

- **Business Logic Processing**
 - The application tier implements organizational rules such as approval workflows, access policies, and data-sharing protocols.
- **API Management**
 - It exposes APIs that allow the presentation tier and external systems to interact with the system. The API gateway manages routing, authentication, and request handling.
- **Authentication and Authorization**
 - Security is enforced at this layer through identity verification and role-based access control (RBAC), ensuring only authorized users can access specific data.
- **Workflow Management**
 - Enterprise processes such as document approvals, notifications, and task assignments are managed here.
- **Integration with External Systems**
 - The application tier integrates with other enterprise systems such as:
 - ERP (Enterprise Resource Planning)
 - CRM (Customer Relationship Management)
 - HR systems
 - Email and notification services
- **Data Processing and Transformation**
 - Raw data retrieved from the data tier is processed, filtered, and formatted before being sent to the presentation tier.

Data Tier

The **Data Tier** is the bottom layer of the Enterprise Information Sharing System. It is responsible for **storing, managing, and retrieving enterprise data**.

This layer ensures that data is **secure, consistent, and available** for use across the organization.

Key Functions

- **Data Storage**

- The data tier stores structured and unstructured data such as:
- User records
- Documents
- Transaction logs
- Metadata

- **Data Retrieval**

- It retrieves data efficiently based on queries from the application tier.

- **Data Integrity and Consistency**

- Mechanisms such as constraints, transactions, and normalization ensure that data remains accurate and consistent.

- **Backup and Recovery**

- Regular backups are maintained to prevent data loss and ensure business continuity.

- **Logging and Auditing**

- All activities are recorded for security, compliance, and monitoring purposes.

- **Search and Indexing**

- Search engines and indexing systems enable fast retrieval of large volumes of enterprise data.

Request Flow in an Enterprise Information Sharing System

In an EISS, information flows across multiple layers in a structured manner. The system follows a **request–response cycle**, where user actions are processed and results are returned efficiently.

Step by Step Flow:

- **User Interaction (Presentation Tier)**

The user initiates a request through a web portal, mobile app, or API client.

- **Request Transmission**

The request is sent to the application tier via APIs (usually HTTPS).

- **Authentication & Authorization (Application Tier)**

The system verifies the user identity and checks permissions.

- **Business Logic Execution**

The application tier processes the request according to enterprise rules.

- **Data Access (Data Tier)**

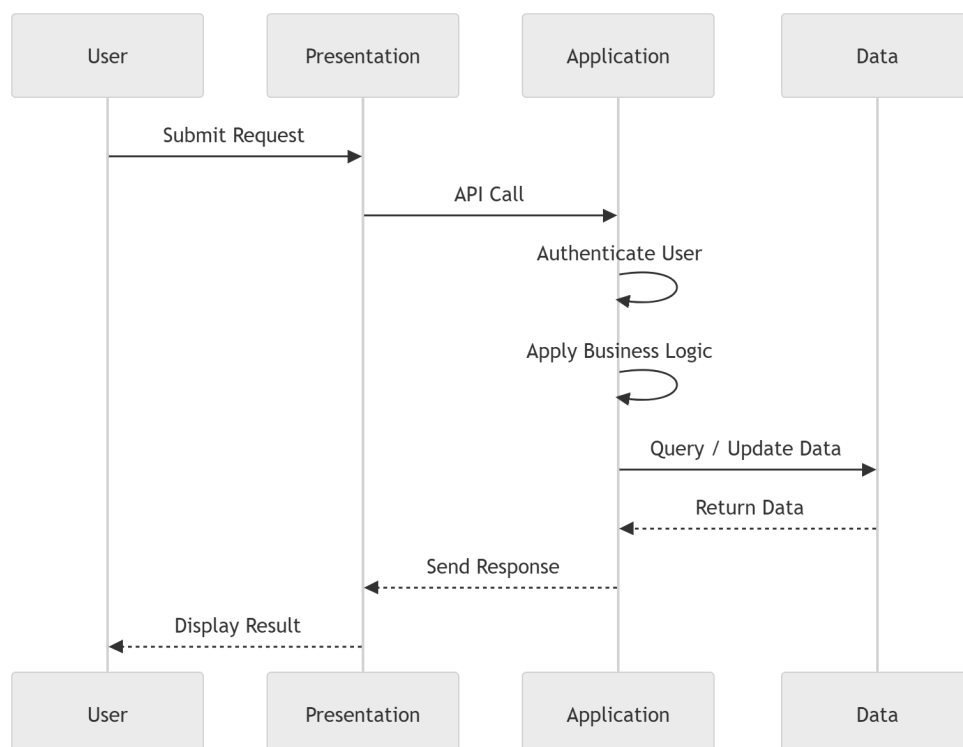
Required data is fetched, updated, or stored.

- **Response Formation**

Processed data is formatted into a response.

- **Response Delivery**

The result is sent back to the presentation tier and displayed to the user.

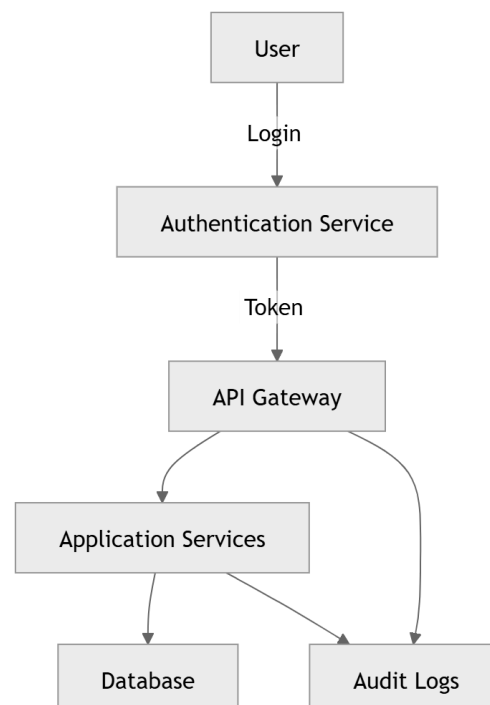


Security in Enterprise Information Sharing Systems

Security is critical in EISS because sensitive enterprise data is shared across multiple users and systems.

Key Security Components

- **Authentication**
 - Username/password
 - Multi-factor authentication (MFA)
 - Single Sign-On (SSO)
- **Authorization**
 - Role-Based Access Control (RBAC)
 - Attribute-Based Access Control (ABAC)
- **Data Encryption**
 - Data in transit → HTTPS / TLS
 - Data at rest → Database encryption
- **API Security**
 - API gateways enforce authentication
 - Rate limiting prevents abuse
 - Token-based access (JWT, OAuth)
- **Auditing and Logging**
 - Tracks user activity
 - Ensures compliance
 - Helps detect anomalies

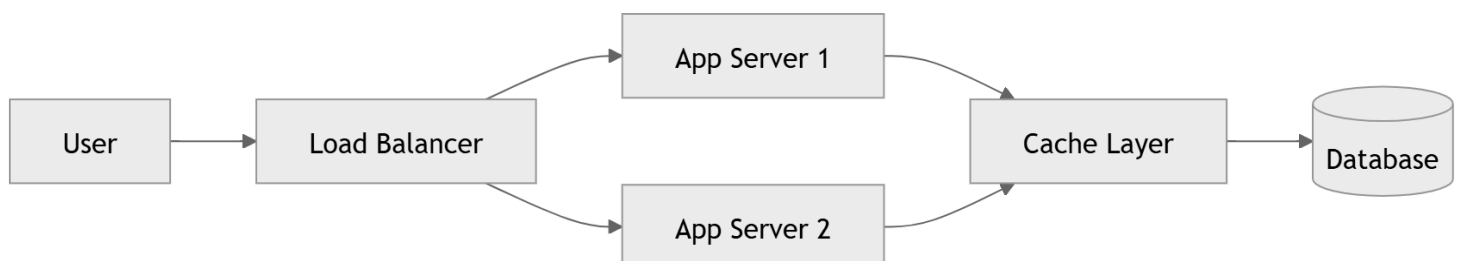


Performance Considerations

Performance ensures that the system can handle large volumes of users and data efficiently.

Key Factors

- **Scalability**
 - Horizontal scaling (adding servers)
 - Vertical scaling (upgrading resources)
- **Load Balancing**
 - Distributes incoming traffic across multiple servers to avoid overload.
- **Caching**
 - Application-level caching
 - CDN caching for static content
- **Database Optimization**
 - Indexing
 - Query optimization
 - Partitioning
- **Asynchronous Processing**
 - Background jobs
 - Message queues (Kafka, RabbitMQ)

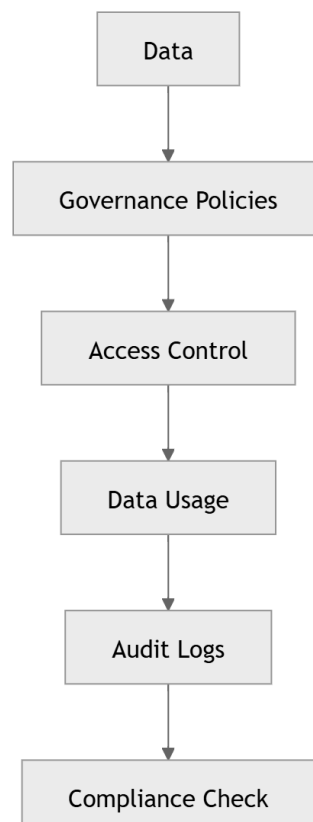


Governance in EISS

Governance ensures that data sharing follows organizational policies, legal regulations, and compliance requirements.

Key Aspects

- Data Governance
 - Data ownership
 - Data classification (sensitive, confidential)
 - Data lifecycle management
- Access Policies
 - Who can access what data
 - Role-based restrictions
- Compliance
 - GDPR
 - HIPAA
 - Industry regulations
- Audit and Accountability
 - Track all system activities
 - Maintain logs for audits



Conclusion

In conclusion, an Enterprise Information Sharing System (EISS) plays a vital role in enabling organizations to efficiently manage, share, and govern information across departments and stakeholders. By adopting a **three-tier architecture**, the system achieves a clear separation of concerns, where the presentation tier handles user interaction, the application tier manages business logic and workflows, and the data tier ensures secure and reliable data storage.

The detailed analysis of these tiers demonstrates how each layer contributes to the overall functionality, scalability, and security of the system. The **request-response flow** highlights the structured movement of information across layers, ensuring that user requests are processed accurately and efficiently. Furthermore, the discussion on **security mechanisms** emphasizes the importance of authentication, authorization, encryption, and auditing in protecting sensitive enterprise data.

In addition, performance considerations such as **scalability, load balancing, caching, and database optimization** ensure that the system can handle increasing workloads while maintaining high responsiveness. Governance aspects further strengthen the system by enforcing **data policies, compliance standards, and accountability**, thereby ensuring responsible and controlled data usage within the organization.

Overall, the integration of architectural design, security practices, performance strategies, and governance policies results in a robust and reliable enterprise system. Such a system not only improves operational efficiency but also enhances decision-making, collaboration, and data-driven processes. Therefore, a well-designed Enterprise Information Sharing System is essential for modern organizations seeking to achieve **scalability, security, and sustainable growth** in a data-driven environment.